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An empirical study on credit card loan delinquency

Hyeongjun Kim[a], Hoon Cho[b], Doojin Ryu[c,*]

[a] Korea Asset Management Corporation (KAMCO), Seoul, Republic of Korea
[b] College of Business, Korea Advanced Institute of Science and Technology (KAIST), Seoul, Republic of Korea
[c] College of Economics, Sungkyunkwan University, 25-2, Sungkyunkwan-ro, Jongno-gu, Seoul 03063, Republic of Korea

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ABSTRACT

Following the Basel II convention, consumer credit default is commonly defined as delinquency beyond a period of 90 days. In this study, rather than considering default as a binary variable, we dissect delinquency states further to investigate default behavior in greater detail. As such, we define three states—no delinquency, delinquency and serious delinquency—and estimate the probabilities of the transitions between states using extensive panel data from Korea, covering a wide range of behavioral information. Our findings have several economic implications. First, the factors that affect delinquency risk can differ from those that affect the transition from delinquency to serious delinquency. Second, the recent increase in the number of seriously delinquent accounts can be attributed to changes in the borrower age distribution. Third, macroeconomic conditions, especially differences in gross domestic product and consumption growth, have led to the recent increase in delinquent accounts. Fourth, the debt-to-income (DTI) ratio has a profound effect on transitions between delinquency states and thus affects both recovery and delinquency. Furthermore, this result is robust to controls for demographic and macroeconomic factors.

1. Introduction

Credit card loans, on which the interest and limits are determined by the borrower's credit card usage, are a primary component of household debt. Unlike a mortgage, this type of debt is unsecured and depends only on the borrower's credit. Credit card loans are widely used in Korea because they are easily accessible and because their interest rates (15–25%) are lower than those of other types of unsecured loans. However, the loan periods of one to three years are short and therefore require relatively large monthly payments. These characteristics (i.e., unsecured and large monthly payments) make assessing their credit risk particularly important.

As Kovrijnykh and Livshits (2017) point out, consumer credit default is not a binary variable. Delinquency, often defined as payments being overdue for more than 60, 90, or 120 days, is considered the first stage of default. If Chapter 7 filings data are not available, consumer credit default is commonly defined as a payment that is three months overdue, following the Basel II convention. In this case, delinquency of less than three months is often regarded as a sign of default because people tend to consider delinquency and default as being caused by the same factors. However, in Korea we observe that the default rate continues to rise even though the delinquency rate has increased more gradually. This is interesting because the trends have not been accompanied by an increase in the probability of delinquency. As a result, a borrower's delinquent behavior might differ from his or her default behavior. Therefore, we hypothesize that the factors that affect delinquency differ from those that affect the transition to default (i.e., serious delinquency). We further hypothesize that a factor exists that reduces the delinquency rate, while increasing the default rate. To test our

* Corresponding author.
E-mail address: sharpjin@skku.edu (D. Ryu).

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hypotheses, we do not solely investigate the default rate. Many studies that focus on default behavior treat the default "state" as a binary variable (e.g., Gross and Souleles, 2002; Agarwal and Liu, 2003; Bellotti and Crook, 2013; Debbaut et al., 2016). In contrast, we dissect the default state into three delinquency states, and then focus on the transitions between them.

In this study, we examine delinquency in credit card loans in Korea by exploiting a unique account-level dataset of the credit card loan market. Using our quarterly observed panel data, we track not only changes in delinquency status, but also changes in account and borrower characteristics, including information related to jobs and income. Consumer credit scoring models predominantly use static information about obligors at the time of application to measure each account's credit risk, and cannot capture changes in default risk over time. Behavioral consumer credit scoring models can capture changes in default risk by using variables that are fixed at the time of estimation, but that change over time. These types of dynamic models allow us to use panel data and variables that change over the estimation period. Ma et al. (2010) and Thomas et al. (2001) argue that dynamic rather than static models should be used in order to fully understand default risk. Furthermore, dynamic models can incorporate macroeconomic variables and time-varying behavioral variables. Macroeconomic variables allow us to capture the empirical implications of default risk in the event of an economic crisis.

This study contributes to the existing literature in several ways. Using unique account-level credit card loan data, we build and investigate a model that measures delinquency risk. Then, we estimate the transition probabilities between delinquency states, investigate which characteristics affect delinquency risk, and examine the features common to accounts that have recovered from delinquency. Because the data are observed quarterly, we also investigate how changes in borrowers' credit ratings, debt-to-income (DTI) ratios and employment status affect delinquency risk. Our results support the view that the delinquency state is viewed as a transitory state, as indicated by Herkenhoff (2012) and Chatterjee (2010), rather than as an absorbing state, as indicated by Drozd and Serrano-Padial (2017). Unlike other studies that define delinquency as payments overdue for more than 90 days, we also examine short-term delinquencies of less than 90 days. Our results show that the factors that cause delinquency are different from those that prolong delinquency. In addition, we use credit card loan data from Korea, which is one of the major emerging economies. As Park (2009) points out, the consumer credit market in Korea has several unique characteristics compared to markets in other developed countries. Our results provide several policy insights, and the study can serve as a reference for readers in other countries interested in household debt.

This paper is organized as follows. Section 2 discusses the consumer credit market in Korea in further detail. Section 3 reviews the current research on credit card loan delinquency risk and develops our hypotheses, while Section 4 describes our data and variables using summary statistics. Section 5 details the methods used for our empirical estimations of the transition probabilities between delinquency states. Section 6 presents the empirical results and, finally, Section 7 concludes the paper.

2. The consumer credit market in Korea

In 2002, Korea experienced an unprecedented credit card crisis. Credit card companies had issued their memberships competitively, which increased the number of credit card holders with poor credit. Credit card usage for purchases and cash advances reached 623 trillion Korean won (KRW) in 2002, with the crisis eventually leading to delinquency for 4 million borrowers. As a result, by 2004, credit card usage had fallen to 357 trillion KRW. The credit card crisis in Korea shares many similarities with the recent financial crisis in the United States. Household debt reached 1 quadrillion KRW in the second quarter of 2013 and, in 2014, credit card usage reached 564 trillion KRW. Credit cards continue to grow in popularity as a method of payment. Fig. 1 shows the total credit card loan amount as a percentage of the gross domestic product (GDP) in Korea. During the credit card crisis in 2002, this measure peaked at over 7% before dropping drastically; recently, however, it has begun increasing again.

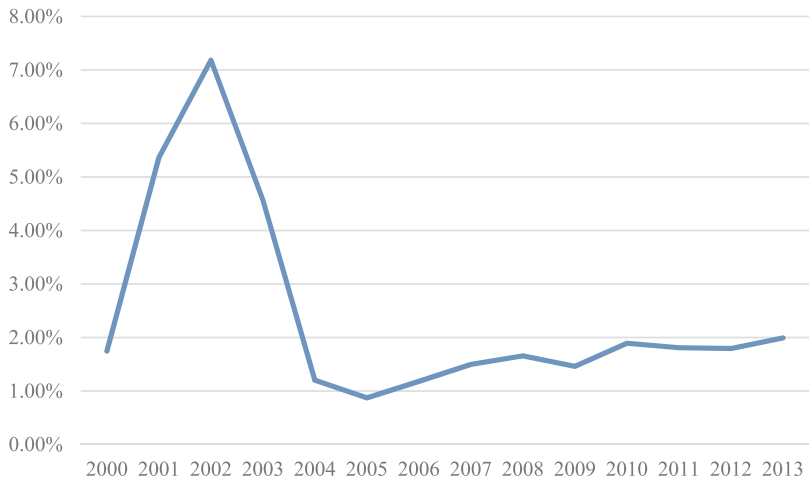

Fig. 1. Credit card loans as a percentage of GDP in Korea.
Notes: This figure shows total credit card loans as a percentage of GDP in Korea. The statistics are measured on an annual basis.

Credit card loans in Korea differ from those in the United States. In Korea, each credit card loan is considered an individual installment loan with a fixed loan period. In contrast, in the United States, a credit card loan is a revolving credit. When a credit card loan payment is overdue, the financial institution imposes a delinquency interest rate by adding a penalty to the original interest rate such that the institution's maximum delinquency interest rate is not exceeded. This penalty may differ according to the delinquency period. Most financial institutions increase their penalties after a payment has been overdue for three months. Some charge an additional penalty when the payment is overdue by more than one month and less than three months. In most cases, the acceleration clause is activated when the payment is overdue by more than one month. Before the acceleration clause is activated, the delinquency interest rate applies only to the overdue principal and interest payment. However, after activation, the delinquency interest rate applies to the overdue payment and to the total outstanding balance. Once the acceleration clause is activated, the financial institution can provisionally seize the debtor's assets and monthly pay account. However, in reality, related lawsuits can take about three months.

In Korea, debtors with heavy debt loads can benefit from a debt-adjustment system. The Credit Counseling and Recovery Service was established in 2002 to support the recovery of credit and finances through an extended payment period, amortization, interest rate reduction and debt reduction. There are two debt-adjustment programs: Individual Workout and Pre-Workout. Individual Workout supports credit and economic recovery through a debt-adjustment program for long-term debtors who have been overdue for three months or more. Pre-Workout is a program to prevent long-term delinquency through preemptive debt adjustment for short-term debtors who have been overdue for less than three months.

3. Literature review and hypothesis development

Consumer credit has been studied widely in the literature, including the forecasting of financial risk. Thomas (2000) surveys credit and behavioral scoring, as well as the primary techniques used. However, empirical works on credit card loans are somewhat limited, mainly because of the inaccessibility of the necessary data. Nevertheless, several studies use unique loan-level datasets. Gross and Souleles (2002) examine credit card accounts by analyzing their credit risk. Emphasizing the role of time-varying factors in models of default, they argue that the propensity to default varies substantially with time.

Agarwal and Liu (2003) investigate the determinants of credit card delinquency by analyzing macroeconomic factors. Their findings show that consumers' propensities for delinquency and bankruptcy vary significantly with macroeconomic fluctuations, and that county unemployment rates have a significant effect on delinquency rates. Agarwal et al. (2003) analyze credit card delinquency by considering whether exemption laws influence the delinquency and bankruptcy behavior of consumers and investors. They find that loose garnishment and property exemption laws result in delinquency and informal bankruptcy, whereas loose homestead and property exemption laws increase formal bankruptcy rates.

Musto and Souleles (2006) analyze unique credit bureau data and calculate the covariance of individual default risk with aggregate consumer default rates. They find that there is significant heterogeneity in covariance risk across market participants and that the amount of credit tends to increase with the credit score, but to decrease with covariance risk. Desai et al. (2014) extend the study of Musto and Souleles (2006), who apply portfolio theory to consumer lending, by analyzing county-level credit outcome betas. By calculating the probability of default using the credit score and actual default and bankruptcy rates, they find that default risk is only weakly correlated with bankruptcy betas.

By analyzing a unique individual- and account-level dataset of US households, Mian and Sufi (2011) find a significant linkage between house prices and household borrowing, which suggests that housing and household leverages explain macroeconomic fluctuations. They also find that house prices do not affect borrowing homogeneously across the population. Homeowners with poor credit and a high tendency to use credit card loans are more sensitive to house prices. Debbaut et al. (2016) examine the relationship between borrower age and the default risk of credit cards in the United States. Their findings show that young borrowers have a relatively low probability of default and are more likely to obtain a mortgage.

Bellotti and Crook (2012) test and compare several models of loss in the case of default using account-level data, including a Tobit model, a decision tree model, a beta model, and fractional logit transformation. The authors claim that ordinary least squares models with macroeconomic variables perform best for forecasting. Then, Bellotti and Crook (2013) propose discrete time survival models of credit card borrower default, and find that dynamic models that include cardholders' behavioral and macroeconomic variables provide significant improvements to the model fit. Leow and Crook (2014) find that debtors' characteristics affect their default and recovery behavior significantly. Dimitrova-Grajzl et al. (2015) conduct a quantitative analysis of consumer credit in Indian country in the United States, revealing relatively limited credit histories. Mazibaş and Tuna (2017) investigate the causes of the recent surge in household debt in Turkey using macroeconomic variables. Kim et al. (2017) show the robust and positive relationship between delinquency probability and private loan amount using consumer credit data from Korea.

Using a theoretical approach, Livshits (2015) gives a comprehensive survey of consumer credit markets and default. Athreya et al. (2018) propose a quantitative model of the renegotiation process using a combined dataset. Kovrijnykh and Livshits (2017) suggest a model with delinquency, renegotiation and bankruptcy when there is an adverse selection problem. Rubaszek and Serwa (2014) propose theoretical and empirical models showing that the amount of household credit is affected by the interest rate spread, income uncertainty and persistence of each individual.

As Kovrijnykh and Livshits (2017) point out, consumer credit default comprises multiple stages rather than being a simple binary event. Here, we define three states of delinquency—no delinquency, delinquency and serious delinquency—and estimate the probabilities of transitions between them. The theoretical models of previous studies suggest that delinquency and repayment follow separate decision-making processes based on future income. We hypothesize that these separate state transitions can be affected by

different factors. Furthermore, there exists a factor that causes delinquency but prevents serious delinquency, or vice versa. These conjectures yield the following two testable hypotheses:

H1. The factors that cause delinquency differ from the factors that prolong delinquency into serious delinquency.

H2. There exists a factor that has opposite effects on each state change.

4. Data

Our data are obtained from an anonymous credit bureau in Korea and consist of 1,590,522 credit card loan accounts issued after 2008:Q3 and observed through 2013:Q2. The accounts are sampled randomly, with a sufficient number to obtain a representative sample. The data include *i*) account characteristics, such as the loan amount, loan period,[1] outstanding loan balance and delinquency state, and *ii*) borrower characteristics, such as age, gender, location of residence, credit rating[2] and DTI ratio. This large dataset also contains each account's quarterly history since being opened, including account and customer characteristics. To prevent re-identification, the data underwent several stages of clustering. The data used in this study are provided on a quarterly basis, with some variables (e.g., age, area, job) clustered into several categories. Owing to its limited accessibility, this unique Korean dataset is rarely analyzed, in spite of its high quality information.

We focus on the delinquency state of each account. Following the Basel II convention, we define three delinquency states: state 0 (no delinquency), where the account is up to date; state 1 (delinquency), where the account is fewer than 90 days delinquent; and state 2 (serious delinquency), where the account is delinquent for 90 days or more. In many studies, state 2 is also known as the default state and is an absorbing state. However, in our study, an account in state 2 can return to state 0 or state 1 if the borrower repays the loan and the interest. Fig. 2 shows the percentages of accounts in our sample in states 1 and 2. Here, panels A and B show the delinquency rate by calendar date and account age, respectively. As shown in the figure, the percentages of delinquent and seriously delinquent accounts have been increasing recently.

Table 1 provides the summary statistics for our sample, including descriptive statistics of the accounts at the time of application and the states of the accounts. We classify the variables into two groups: application variables and behavioral variables. The application variables are static and take the values in the quarter in which a loan is issued. The variables include loan characteristics, such as the amount and the period, as well as borrower characteristics, such as gender, age, area, credit rating, DTI ratio and job status.

Because the account information is observed quarterly, we can track changes in the behavioral variables. Here, we include changes in a borrower's credit rating, DTI ratio and job status as explanatory variables. We can also observe the loan balances. As such, we measure the outstanding loan balance ratio as the ratio between the current and the original loan balance. Changes in this ratio are also included in our model as an explanatory variable. To capture economic status, we include six macroeconomic variables in the model: the unemployment rate, term spread, credit spread, KRW–dollar exchange rate, GDP growth and consumption growth. Fig. 3 shows the macroeconomic variables during the sample period in relation to delinquency states 1 and 2.

Table 2 shows the transition matrix for the delinquency states. State 2, in which an account is more than 90 days delinquent, is not an absorbing state. A total of 52.90% of accounts in state 2 recover, whereas 46.07% remain in state 2. Therefore, we cannot use default probability analysis models, such as survival analysis models, because they assume that the default state is an absorbing state. Because our data are observed on a quarterly basis, the movements between states are not sequential. An account that was issued in quarter τ or that was in state 0 in quarter $\tau - 1$ can only move to state 1 or remain in state 0 in quarter τ. However, an account in state 2 in quarter $\tau - 1$ can recover to state 0 or state 1 in quarter τ.[3]

5. Methodology

In the literature, the multinomial logit model specification is often used when there are multiple potential outcomes (e.g., Agarwal et al., 2003; Pennington-Cross, 2010). An account can only be in one state at any given moment, and thus the multinomial logit model can also be referred to as a competing risk model. If there are J possible outcomes, the probability of a specific outcome j ($j = 0, 1, \ldots, J - 1$) is given as follows:

$$\Pr(y_i = j \mid \mathrm{Xi}) = e^{X_i\beta_j} / \sum_{k=0}^{J-1} e^{X_i\beta_k} \tag{1}$$

where X_i is the vector of explanatory variables for the i-th loan, and β_j is the vector of regression coefficients corresponding to outcome j.

The parameters in Eq. (1) are estimated by maximizing the following log-likelihood function:

[1] The account characteristics of credit card loans in Korea include the loan period because the loan is considered as an individual installment loan with a fixed loan period.

[2] Credit ratings may create a multicollinearity problem with other variables. To resolve this problem, we regress credit rating on loan/borrower characteristic variables and use the residual as an explanatory variable that contains the unique information provided by the credit rating. However, we find that neither the signs of the parameters nor their significance levels change for any important variables. Therefore, for simplicity, we use the original credit rating as the explanatory variable.

[3] A movement from state 2 to state 1 means that the borrower is again in arrears after recovering from delinquency.

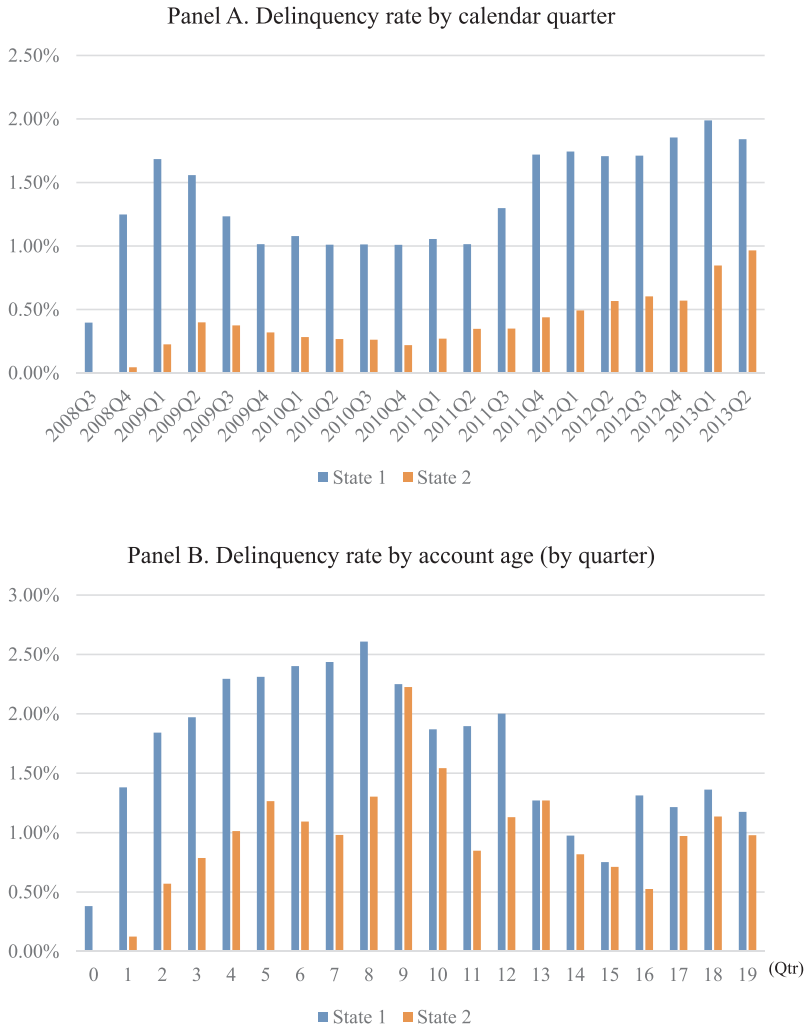

Fig. 2. Delinquency rates of credit card loans in Korea.
Notes: This figure shows the delinquency rates of credit card loans in Korea. State 1 indicates that an account has been delinquent for fewer than 90 days, and state 2 indicates that an account has been delinquent for 90 days or more, which we call serious delinquency. Panels A and B show the delinquency rate by calendar date and by account age (by quarter), respectively.

$$\log \mathrm{L} = \sum_{i} \sum_{j=0}^{J-1} I(y_i = j) \log \left(\Pr(y_i = j \,\middle|\, \mathrm{Xi})\right), \tag{2}$$

where $I(\cdot)$ is an indicator function and is equal to one if the outcome of the i-th loan is j.

The chief criticism of applying the multinomial logit model in this case is that the loan termination risk setting does not satisfy the required independence of irrelevant alternatives (IIA) assumption, because default and prepayment are substitutes. The ratio of the probabilities for any two alternatives can be expressed as

$$Pr(y_i = m \,\middle|\, Xi) \,/\, Pr(y_i = n \,\middle|\, Xi) = \frac{e^{X_i\beta_m} / \sum_{k=0}^{J-1} e^{X_i\beta_m}}{e^{X_i\beta_n} / \sum_{k=0}^{J-1} e^{X_i\beta_n}} = e^{X_i\beta_m - X_i\beta_n} \tag{3}$$

The multinomial logit model assumes that this odds ratio for alternatives m and n does not depend on any other alternatives. The Hausman–McFadden test is widely used to test whether the IIA assumption is violated. Thus, following Hausman and McFadden (1984), we run a multinomial logit for all possible J outcomes and estimate the unrestricted parameter $\hat{\beta}_U$. Then, we run another multinomial logit for $J - 1$ outcomes and estimate the restricted parameter $\hat{\beta}_R$. If the number of independent variables is k, then the Hausman–McFadden test statistic is

$$(\hat{\beta}_U - \hat{\beta}_R)' [\hat{V}_U - \hat{V}_R]^{-1} (\hat{\beta}_U - \hat{\beta}_R) \sim \chi^2_{((J-2)k)} \tag{4}$$

Table 1
Descriptive statistics.

		At application	State 0	State 1	State 2
N		1,590,522	5,329,681	72,842	28,879
Account age (qtr.)			2.05	3.00	4.00
			(2.26)	(2.28)	(2.37)
Application variables					
Loan amount (thousands of KRW)		3400.07	3649.29	4270.03	4493.59
		(3061.12)	(3071.26)	(3158.36)	(3711.80)
Loan period (months)		15.11	18.30	21.20	22.82
		(10.05)	(11.32)	(12.32)	(17.37)
Gender	Male	48.70%	48.32%	56.01%	59.98%
	Female	51.30%	51.68%	43.99%	40.02%
Age		41.54	42.27	41.73	43.23
		(9.65)	(9.65)	(9.86)	(9.75)
Age group	20 ≤ age < 30	9.71%	8.20%	9.86%	7.12%
	30 ≤ age < 35	16.68%	15.48%	16.63%	13.81%
	35 ≤ age < 40	19.71%	19.13%	18.70%	17.51%
	40 ≤ age < 45	18.36%	18.76%	18.38%	18.18%
	45 ≤ age < 50	14.15%	14.78%	14.26%	16.18%
	50 ≤ age < 55	10.67%	11.65%	10.94%	13.81%
	55 ≤ age	10.73%	12.00%	11.25%	13.40%
Area	Seoul	57.89%	57.93%	59.82%	61.81%
	Non-Seoul	42.11%	42.07%	40.18%	38.19%
Credit rating		6.14	6.22	8.70	9.90
		(1.07)	(1.19)	(1.02)	(0.74)
DTI ratio		34.85%	40.96%	273.82%	523.60%
		(50.82%)	(57.58%)	(285.05%)	(344.81%)
Job	Salaried employee	45.63%	44.14%	39.85%	40.92%
	Self-employed	12.61%	13.00%	14.37%	16.36%
	Others	28.24%	29.44%	31.38%	33.24%
	No information	13.52%	13.41%	14.41%	9.48%
Behavioral variables					
Outstanding loan balance ratio		0.88	0.63	0.62	0.65
		(0.26)	(0.35)	(0.29)	(0.32)
Change in credit rating			0.04	1.57	2.57
			(0.80)	(1.34)	(1.59)
Change in DTI ratio			7.98%	206.30%	409.92%
			(50.35%)	(277.06%)	(370.19%)
Change in job status:					
(at application)	(now)				
Salaried employee	Self-employed		0.56%	0.94%	1.59%
Salaried employee	Others		1.82%	2.67%	3.65%
Salaried employee	No information		0.84%	1.20%	1.12%
Self-employed	Salaried employee		0.80%	1.36%	2.01%
Self-employed	Others		0.78%	1.33%	1.90%
Self-employed	No information		0.34%	0.59%	0.64%
Others	Salaried employee		2.49%	4.63%	7.05%
Others	Self-employed		0.82%	1.43%	2.40%
Others	No information		1.11%	1.92%	1.95%
No information	Salaried employee		0.32%	0.58%	0.59%
No information	Self-employed		0.11%	0.20%	0.18%
No information	Others		0.34%	0.49%	0.49%

Notes: This table presents the descriptive statistics for our data. Our data include 1,590,522 credit card loan accounts issued after 2008:Q3 and observed quarterly through 2013:Q2. For the dependent variables, three delinquency states are defined: state 0 (no delinquency), which is an up-to-date account; state 1 (delinquency), where the account has been delinquent for fewer than 90 days; and state 2 (serious delinquency), in which the account has been delinquent for 90 days or more. The application variables are static variables, taking the values in the quarter in which the loans are issued. The behavioral variables are time-varying variables and track changes in the explanatory variables. Changes in the credit rating, DTI ratio and job status are defined as the differences in these values between the time of the account application and the time of observation.

where $\hat{V_U}$ and $\hat{V_R}$ represent the variance–covariance matrices of the unrestricted and restricted models, respectively.

6. Empirical results

Table 3 presents the regression results of the multinomial logit estimation and describes the types of accounts that are more likely to move from the normal state to delinquency. Panel A of Table 3 reports the results for accounts in the quarter in which they were first issued and estimates the probability of a transition to state 1 (i.e., the delinquency state). Panel B shows the results for accounts that were not delinquent in the previous quarter and estimates the probability of a transition from a non-delinquent to a delinquent

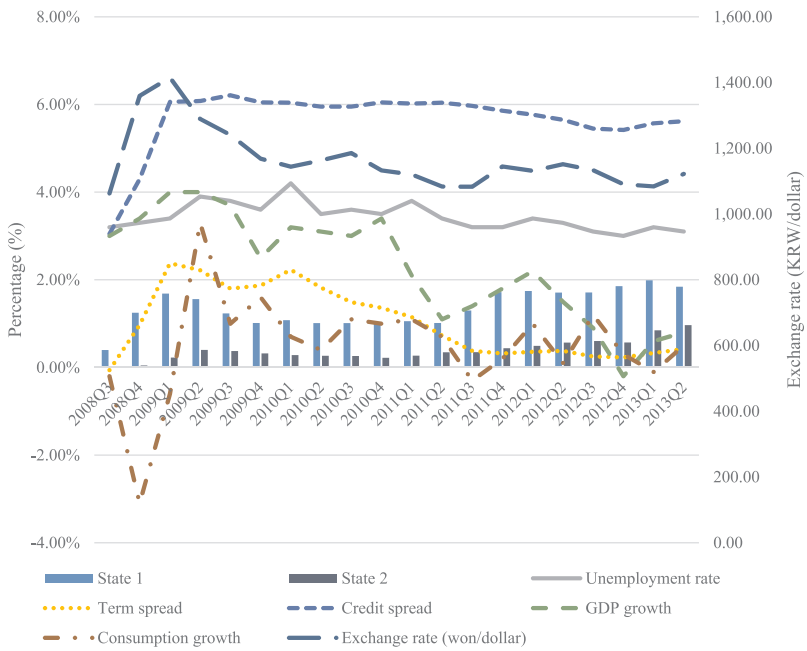

Fig. 3. Macroeconomic variables during the sample period.
Notes: This figure shows the macroeconomic variables that are widely used to capture the economic status during the sample period. States 1 and 2 denote delinquency and serious delinquency (i.e., default), respectively, for credit card loans.

Table 2
Transition probabilities between delinquency states.

From \ To	State 0	State 1	State 2
Loan issue	99.62%	0.38%	–
State 0	98.34%	1.66%	–
State 1	49.42%	17.77%	32.81%
State 2	52.90%	1.03%	46.07%

Notes: This table reports the transition probabilities between the three delinquency states: state 0 (no delinquency), which is an up-to-date account; state 1 (delinquency), where the account has been delinquent for fewer than 90 days; and state 2 (serious delinquency), in which the account has been delinquent for 90 days or more. Because the data are observed on a quarterly basis, an account cannot transition to state 2 if it was just issued or if it was in state 0 in the previous quarter.

state. In this case, we ignore a transition from state 0 to state 2 because the data are observed quarterly; thus, by definition, accounts in state 0 cannot move directly to state 2. Only panel B of Table 3 includes the account age (in quarters) because all accounts are the same age in the quarter in which they are first issued. The account age captures the seasoning effect of a loan. The results show that an account's risk of delinquency decreases as it ages.

For the application variables, accounts with larger loan amounts and shorter loan periods have a higher delinquency risk. Larger loans with a shorter maturity are associated with higher monthly payments, which can be a burden for borrowers. Women have a lower delinquency risk than men. The coefficients for every age group are significantly negative, which implies that borrowers under 30 years of age are riskier than borrowers in the other age groups.[4] The non-Seoul area dummy is not statistically significant in panel A of Table 3, but significant in panel B. Borrowers with lower credit ratings have a greater delinquency risk.[5] A higher DTI ratio implies a higher delinquency risk. The borrower's job status also affects the delinquency risk. Panel A shows that in the quarter in which a loan is issued, self-employed borrowers do not have a significantly lower delinquency risk than salaried employees, who constitute the reference group. However, self-employed borrowers develop a higher risk of delinquency over time. Borrowers with other types of jobs or with no job status information have a higher delinquency risk than salaried employees.[6] Then, accounts with

[4] In order to examine the statistical differences between the dummy variables, we run Wald tests using each estimate. In Panel A, the group of borrowers aged between 30 and 35 years is statistically different from the group aged over 45. The group aged between 35 and 40 is statistically different from the group aged over 50. In Panel B, each age group is statistically different from the other groups, except for the group aged over 55.

[5] In Korea, a borrower's credit rating ranges from one to ten, with a lower numerical score indicating a better credit history. Thus, a lower credit rating means a higher numerical score.

[6] The results of the Wald tests show that each job status group differs from the other groups.

Table 3
Multinomial logit model of loan delinquency (from issue and state 0).

		Panel A: From loan issue to state 1			Panel B: From state 0 to state 1		
		Coeff.		Std. error	Coeff.		Std. error
Constant		−23.3664	***	0.6425	−19.8007	***	0.3319
Account age (qtr.)					0.0221	***	0.0026
Application variables							
Log of loan amount		0.2978	***	0.0176	0.4247	***	0.0064
Loan period		−0.0209	***	0.0012	−0.0173	***	0.0004
Women dummy		−0.2323	***	0.0294	−0.2043	***	0.0098
Age group (reference: age < 30)							
30 ≤ age < 35		−0.1986	***	0.0541	−0.1723	***	0.0174
35 ≤ age < 40		−0.2633	***	0.0529	−0.3111	***	0.0172
40 ≤ age < 45		−0.2367	***	0.0526	−0.3805	***	0.0175
45 ≤ age < 50		−0.4114	***	0.0575	−0.4732	***	0.0188
50 ≤ age < 55		−0.4404	***	0.0625	−0.5401	***	0.0206
55 ≤ age		−0.4557	***	0.0628	−0.4094	***	0.0205
Non-Seoul area dummy		0.0350		0.0292	0.0486	***	0.0098
Credit rating		0.8950	***	0.0121	0.9121	***	0.0050
DTI ratio		0.0062	***	0.0001	0.0024	***	0.0001
Job status (reference: salary employed)							
Self-employed		−0.0024		0.0458	0.0818	***	0.0169
Others		0.1165	***	0.0355	0.1844	***	0.0131
No information		0.3664	***	0.0427	0.4195	***	0.0150
Behavioral variables							
Outstanding loan balance ratio		0.3872	***	0.0566	1.2602	***	0.0185
Change in area:							
(from)	(to)						
Non-Seoul	Seoul				0.0888	*	0.0462
Seoul	Non-Seoul				0.0227		0.0481
Change in credit rating					0.7016	***	0.0049
Change in DTI ratio					0.0075	***	0.0000
Change in job:							
(from)	(to)						
Salary employed	Self-employed				0.0903	**	0.0424
Salary employed	Others				0.1310	***	0.0246
Salary employed	No information				0.4458	***	0.0624
Self-employed	Salary employed				−0.0752		0.0541
Self-employed	Others				0.1229	***	0.0430
Self-employed	No information				0.2915	***	0.1039
Others	Salary employed				−0.0549	*	0.0306
Others	Self-employed				0.0110		0.0431
Others	No information				0.2112	***	0.0653
No information	Salary employed				−0.1885	***	0.0444
No information	Self-employed				−0.0237		0.0619
No information	Others				0.0201		0.0355
Macroeconomic variables							
GDP growth		−0.1034	***	0.0272	−0.1037	***	0.0106
Consumption growth		0.0648	***	0.0170	0.0621	***	0.0066
Unemployment rate		0.6508	***	0.1083	0.0055		0.0381
Term spread		−0.6094	***	0.0532	−0.1454	***	0.0178
Credit spread		0.0555	*	0.0328	−0.0464	**	0.0236
Exchange rate (KRW/dollar)		0.0048	***	0.0003	0.0027	***	0.0001
$-2lnL$		58,250.99			433,088.51		
N		1,590,331			3,774,504		

Notes: This table reports the regression results of the multinomial logit model. Three delinquency states are defined as dependent variables: state 0 (no delinquency), which is an up-to-date account; state 1 (delinquency), where the account has been delinquent for fewer than 90 days; and state 2 (serious delinquency), in which the account has been delinquent for 90 days or more. The reference state is state 0 (no delinquency). There is no transition to state 2 in this table because the data are observed quarterly. Panel A shows the results of the regression for accounts in the quarter they were issued, and panel B shows the results of the regression for accounts that were in state 0 in the previous quarter. The superscripts ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The sample period is from 2008:Q3 to 2013:Q2.

larger remaining loan balances are more likely to be delinquent.

We also include behavioral variables to capture the effects of changes in borrower status after a loan application. An extensive body of literature has investigated the effects of credit ratings and job status on delinquency and default risk. However, changes in these variables at the individual level have been tested less frequently because, in general, loan data only include the borrower's information at the time of application. The behavioral variables have several interesting features in the investigation of delinquency

probabilities. The outstanding loan balance ratio is the only behavioral variable shown in panel A of Table 3 because no other changes are observed during the quarter of issuance. The coefficients on the outstanding loan balance ratio are positive and significant. Accounts with larger remaining loan balances have a higher delinquency risk. Moving from a non-Seoul area to the Seoul area raises a borrower's delinquency risk. Changes in the credit rating also affect the delinquency risk. If a borrower's current credit rating is lower than it was upon application, then that borrower's account has a higher delinquency risk than those of borrowers with the same credit rating. Growth in the DTI ratio also increases the delinquency risk. Borrowers who experience a positive DTI ratio change, that is, a growth in debt or a decrease in income, have a higher risk of delinquency than do other borrowers under similar conditions in the same quarter. Interestingly, the delinquency risk increases if salaried employees transition to other job categories, or if self-employed borrowers transition to any job category other than salaried jobs. However, the delinquency risk decreases when borrowers in other job categories and those with no job information become salaried employees.

Macroeconomic variables also affect the delinquency risk, which diminishes when GDP growth is positive and when consumption growth is negative. We can interpret GDP growth as household income, and consumption growth as household expenses. Thus, as income increases (decreases) and expenses decrease (increase), the delinquency risk decreases (increases). The coefficient of the unemployment rate is positive and significant in panel A of Table 3, but not statistically significant in panel B. These results follow from including behavioral variables in the model, especially variables related to job changes. Increasing the term spread decreases the delinquency risk. The credit spread has positive effects on the delinquency risk in the 90 days after loan issuance, but negative effects thereafter. An increase in the KRW–dollar exchange rate also increases the delinquency risk. In other words, the risk of delinquency increases as the US dollar becomes more expensive relative to the KRW.

Table 4 shows the results of the multinomial logit regression analysis of delinquent accounts (i.e., state 1). The table shows the types of accounts that are more likely to recover from delinquency and those that are more likely to become seriously delinquent. Panel A estimates the probability of a transition from state 1 to state 0, and panel B estimates the probability of a transition from state 1 to state 2. A transition to state 0 means the account has recovered, and a transition to state 2 means the account has become seriously delinquent, which many studies define as the default state. In this model, state 1 is used as the reference state.

The coefficients on account age show that seasoned accounts are more likely to become seriously delinquent, whereas unseasoned accounts are more likely to recover. Small loans can easily transition to other states. Accounts with longer loan periods are more likely to recover, whereas shorter loan periods tend to lead to serious delinquency. The indicator for female borrowers is not statistically significant in panels A or B of Table 4. Although there are no gender differences when the accounts are delinquent, female borrowers do have a lower delinquency risk when they are not delinquent. Borrowers over 35 years of age are more likely to recover from delinquency, whereas borrowers over 45 are more likely to become seriously delinquent if they are delinquent already. This finding is interesting because, as shown in Table 3, borrowers who are older than 45 have a lower delinquency risk when in the normal, non-delinquent state. The borrower's residential location does not affect the transition probabilities in this model. Borrowers with high credit ratings are more likely to recover from delinquency and less likely to become seriously delinquent. A higher DTI ratio increases the probability of a transition to serious delinquency. If there is no job information about the borrower, then the account is less likely to recover.

We next consider the behavioral variables. The coefficients on the outstanding loan balance ratio imply that a larger remaining loan balance decreases the probability of delinquency recovery and increases that of serious delinquency. Borrowers who move to Seoul are more likely to recover from delinquency. A higher credit rating (i.e., a worse credit history) increases the probability of serious delinquency and decreases the probability of being in state 0. Changes in a borrower's DTI ratio affect the delinquency risk. Certain job status changes also affect the transition probabilities. Salaried borrowers are more likely to become seriously delinquent if their job categories change, and less likely to recover from delinquency if they no longer have any job information. If self-employed borrowers become salaried employees, they are more likely to recover from delinquency.

Finally, we consider the macroeconomic variables. Positive GDP growth decreases the probability of a transition to state 2. Positive consumption growth decreases the probabilities of transitions to state 0 and state 2. The unemployment rate is not statistically significant in either panel A or panel B of Table 4, which we assume is due to the inclusion of job change variables. An increase in the term spread increases the probability of serious delinquency, whereas an increase in the credit spread increases the probability of recovery from delinquency. The exchange rate is not statistically significant in either panel A or panel B.

Table 5 presents the results of the multinomial logit regression analysis of accounts in state 2 (i.e., serious delinquency) and reports the types of accounts that are more likely to recover from serious delinquency. Panel A of Table 5 estimates the probability of a transition from state 2 to state 0, and panel B estimates the probability of a transition from state 2 to state 1. A transition to state 0 means the account has recovered, while a transition to state 1 means that the account has recovered, but the borrower is again in arrears. Therefore, the coefficients of the explanatory variables, when significant, have similar signs for both states. Here, state 2 is used as the reference state.

The coefficients of account age indicate that unseasoned accounts are more likely to recover from delinquency, whereas accounts with larger loan amounts and longer loan periods are more likely to recover from serious delinquency. Borrowers between 30 and 35 years of age have a higher probability of recovery than the reference group, whereas borrowers between 45 and 50 and those older than 55 are more likely to remain seriously delinquent. Borrowers who live in the Seoul area are more likely to recover and borrowers with a higher credit rating are more likely to repay their delinquent loans. Accounts with a lower DTI ratio are more likely to recover. However, if a seriously delinquent borrower has no job information, the probability of recovery increases, which we believe is driven by another person (e.g., parents or a partner) repaying the loan rather than by the borrower doing so.

Borrower behavior also affects the probability of recovery from delinquency. A smaller remaining loan balance increases the probability of recovery. A drop in the credit rating increases the probability of transitioning from state 2 to state 1. Borrowers who

Table 4
Multinomial logit model of loan delinquency (from state 1).

		Panel A: From state 1 to state 0			Panel B: From state 1 to state 2		
		Coeff.		Std. error	Coeff.		Std. error
Constant		4.0526	***	1.0505	−2.9770	**	1.3320
Account age (qtr.)		−0.3483	***	0.0090	0.2837	***	0.0114
Application variables							
Log of loan amount		−0.1305	***	0.0229	−0.4263	***	0.0273
Loan period		0.0490	***	0.0016	−0.0353	***	0.0019
Women dummy		0.0055		0.0278	−0.0105		0.0344
Age group (reference: age < 30)							
30 ≤ age < 35		0.0749		0.0496	0.0455		0.0630
35 ≤ age < 40		0.1160	**	0.0487	0.0815		0.0618
40 ≤ age < 45		0.1325	***	0.0497	0.0910		0.0625
45 ≤ age < 50		0.1964	***	0.0533	0.2455	***	0.0660
50 ≤ age < 55		0.1821	***	0.0585	0.2563	***	0.0714
55 ≤ age		0.1411	**	0.0595	0.3978	***	0.0725
Non-Seoul area dummy		−0.0265		0.0279	−0.0129		0.0345
Credit rating		−0.2423	***	0.0165	0.6465	***	0.0212
DTI ratio		−0.0013	***	0.0001	0.0030	***	0.0001
Job status (reference: salary employed)							
Self-employed		−0.0498		0.0492	0.0518		0.0587
Others		−0.0176		0.0379	0.0195		0.0468
No information		−0.1394	***	0.0433	−0.0339		0.0560
Behavioral variables							
Outstanding loan balance ratio		−3.7852	***	0.0679	3.8040	***	0.0853
Change in area:							
(from)	(to)						
Non-Seoul	Seoul	0.2346	*	0.1214	0.0216		0.1493
Seoul	Non-Seoul	0.1132		0.1323	0.1206		0.1579
Change in credit rating		0.0186		0.0152	0.7854	***	0.0203
Change in DTI ratio		−0.0016	***	0.0001	0.0033	***	0.0001
Change in job:							
(from)	(to)						
Salary employed	Self-employed	0.0736		0.1133	0.0261		0.1336
Salary employed	Others	0.0902		0.0666	0.1385	*	0.0773
Salary employed	No information	−0.3465	**	0.1417	0.0481		0.1792
Self-employed	Salary employed	0.3903	***	0.1453	0.0524		0.1656
Self-employed	Others	−0.0496		0.1119	−0.0681		0.1292
Self-employed	No information	−0.0369		0.2445	−0.4737		0.3182
Others	Salary employed	0.0232		0.0849	0.1671	*	0.0993
Others	Self-employed	0.1142		0.1183	−0.0063		0.1348
Others	No information	0.0842		0.1692	−0.1012		0.2281
No information	Salary employed	0.0441		0.1108	0.1404		0.1386
No information	Self-employed	0.2179		0.1734	0.3873	*	0.2028
No information	Others	0.0857		0.0952	0.2208	*	0.1171
Macroeconomic variables							
GDP growth		0.0210		0.0321	−0.1525	***	0.0412
Consumption growth		−0.0363	*	0.0203	−0.0492	*	0.0253
Unemployment rate		0.1353		0.1114	−0.1431		0.1436
Term spread		−0.0028		0.0493	0.3861	***	0.0626
Credit spread		0.3159	***	0.0920	0.1074		0.1138
Exchange rate (KRW/dollar)		0.0001		0.0004	0.0002		0.0006
$-2lnL$		62,094.17					
N		48,424					

Notes: This table reports the regression results of a multinomial logit model using accounts that were in state 1 (i.e., fewer than 90 days delinquent) in the previous quarter. Panel A shows the results for the transition to state 0 (no delinquency), which denotes recovery, and panel B shows the results for the transition to state 2 (serious delinquency). The reference state is state 1, which means the account remains in the same state as in the previous quarter. The superscripts ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

experience a decrease in their DTI ratio are more likely to recover from delinquency. Several job transition indicators are also significant. With regard to the macroeconomic variables, the probability of recovery from delinquency diminishes when GDP growth is positive and when consumption growth is negative. A lower unemployment rate increases the probability of account recovery. Wider term spreads, narrower credit spreads, and higher exchange rates increase the probability of a transition to state 0.

The results in Tables 3–5 support our hypotheses and have several economic implications. First, the factors that affect the transition from the normal state to the delinquent state can differ from those that affect the transition from the delinquent state to the seriously delinquent state. For example, Table 3 shows that women have a lower delinquency risk than men. However, if an account

Table 5
Multinomial logit model of loan delinquency (from state 2).

		Panel A: From state 2 to state 0			Panel B: From state 2 to state 1		
		Coeff.		Std. error	Coeff.		Std. error
Constant		0.7628		1.8461	−14.3784	**	6.2425
Account age (qtr.)		−0.1579	***	0.0097	−0.1208	***	0.0381
Application variables							
Log of loan amount		0.1158	***	0.0253	0.4267	***	0.1047
Loan period		0.0223	***	0.0010	0.0062	*	0.0032
Women dummy		0.0130		0.0421	0.2728	*	0.1552
Age group (reference: age < 30)							
30 ≤ age < 35		0.1917	**	0.0858	−0.3543		0.2950
35 ≤ age < 40		0.0675		0.0833	−0.2327		0.2701
40 ≤ age < 45		0.0382		0.0836	−0.2738		0.2722
45 ≤ age < 50		−0.1836	**	0.0849	−0.3967		0.2764
50 ≤ age < 55		0.1154		0.0900	−0.4919		0.3308
55 ≤ age		−0.2627	***	0.0918	−0.5139		0.3139
Non-Seoul area dummy		−0.1078	**	0.0422	0.0284		0.1550
Credit rating		−0.1512	***	0.0334	0.0398		0.0738
DTI ratio		−0.0022	***	0.0001	−0.0054	***	0.0007
Job status (reference: salary employed)							
Self-employed		−0.0020		0.0684	−0.4042		0.2987
Others		0.0685		0.0575	0.4060	**	0.1982
No information		0.3528	***	0.0808	−0.5052		0.3506
Behavioral variables							
Outstanding loan balance ratio		−4.0130	***	0.0636	−1.3337	***	0.2100
Change in area:							
(from)	(to)						
Non-Seoul	Seoul	−0.3270	*	0.1685	−1.3147		1.0164
Seoul	Non-Seoul	−0.1885		0.1796	−0.2411		0.7328
Change in credit rating		0.0430		0.0304	−0.2027	***	0.0548
Change in DTI ratio		−0.0029	***	0.0001	−0.0034	***	0.0004
Change in job:							
(from)	(to)						
Salary employed	Self-employed	−0.1911		0.1446	0.6651		0.4449
Salary employed	Others	−0.1530	*	0.0835	0.4000		0.2998
Salary employed	No information	0.1283		0.2327	−12.0497		335.3000
Self-employed	Salary employed	−0.4052	***	0.1535	0.6250		0.5367
Self-employed	Others	−0.4226	***	0.1343	0.2473		0.5752
Self-employed	No information	0.8461	*	0.4383	−11.1431		591.9000
Others	Salary employed	−0.0007		0.1102	−0.2391		0.4142
Others	Self-employed	−0.2227		0.1539	−12.0543		186.4000
Others	No information	−0.1925		0.2528	−0.7087		1.0298
No information	Salary employed	−0.0533		0.2027	1.6254	***	0.5761
No information	Self-employed	−0.2745		0.2490	−10.5993		215.8000
No information	Others	−0.1262		0.1514	0.2278		0.6716
Macroeconomic variables							
GDP growth		−0.1672	***	0.0567	−0.0965		0.2057
Consumption growth		0.1488	***	0.0414	−0.0121		0.1331
Unemployment rate		−0.9853	***	0.1826	−1.0039		0.6751
Term spread		−0.1741	**	0.0775	0.0161		0.2791
Credit spread		1.4631	***	0.1881	1.3414	*	0.6959
Exchange rate (KRW/dollar)		−0.0034	***	0.0008	0.0023		0.0025
$-2lnL$		17,345.72					
N		18,143					

Notes: This table reports the regression results of a multinomial logit model using accounts that were in state 2 (i.e., delinquent for more than 90 days) in the previous quarter. Panel A shows the results for a transition to state 0 (no delinquency), denoting recovery, and panel B shows the results of a transition to state 1 (the delinquency state), which indicates that the account recovered and then became delinquent again. The reference state is state 2, which means the account remained in the same state. The superscripts ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

is already in the delinquency state, gender does not affect the transition probabilities (see Tables 4 and 5). Job status also affects the probability of a transition from state 0 to state 1. Although salaried employees have a lower delinquency risk than other borrowers, they do not necessarily have a lower risk of becoming seriously delinquent. This finding is consistent with the result of Leow and Crook (2014).

Second, the recent increase in the number of seriously delinquent accounts (see Fig. 2) may be due to the increase in the number of older borrowers. We hypothesize that the phenomenon illustrated in Fig. 2 can be explained by a factor that reduces the

Table 6
Distribution of borrower ages at account application.

Quarter	Age > 45
2008Q3	31.06%
2008Q4	28.18%
2009Q1	29.92%
2009Q2	29.55%
2009Q3	30.97%
2009Q4	31.72%
2010Q1	33.34%
2010Q2	32.61%
2010Q3	34.26%
2010Q4	34.27%
2011Q1	35.68%
2011Q2	35.04%
2011Q3	36.78%
2011Q4	36.52%
2012Q1	39.58%
2012Q2	39.09%
2012Q3	40.19%
2012Q4	40.08%
2013Q1	42.02%
2013Q2	41.71%

Notes: This table reports the proportion of borrowers over the age of 45 at the time of account application, by quarter. This measure is scaled by the number of new loans in the relevant quarter.

delinquency rate, while increasing the default rate; borrowers over 45 satisfy this condition perfectly. As shown in Tables 3 and 4, borrowers who are older than 45 are less likely to be delinquent. However, if they are in state 1, they are more likely to become seriously delinquent. This characteristic also holds for recovery transitions. Panel A in Table 4 shows that borrowers over 45 are more likely to move to the normal state from the delinquency state than borrowers under 30. However, if the former are in the serious delinquency state (see Table 5), these borrowers are less likely to recover. An increase in the size of this group can lead to an increase in the number of seriously delinquent accounts compared to that of merely delinquent accounts. We confirm this result in Table 6, which verifies that the percentage of borrowers older than 45 has recently increased. This result can be explained by the fact that, in most cases, borrowers over 45 are responsible for their family's livelihood.[7]

In addition, the recent increase in the number of accounts in the delinquency state can be attributed to macroeconomic variables. Fig. 3 shows that the delinquency rate has been increasing since the second half of 2011. Several significant variables affect delinquency, but the difference between GDP growth and consumption growth has narrowed in the past few years. Comparing this change with other variable changes, we conclude that the macroeconomic climate has caused an increase in delinquency.

Finally, the results all show that the DTI ratio plays an important role in the transitions between delinquency states. Both the initial level and subsequent changes in a borrower's DTI ratio increase the probability of moving to delinquency from the normal state, and the probability of moving to serious delinquency from the delinquent state. The DTI ratio also has a significant impact on the probability of recovering from a delinquent or seriously delinquent state. This result is robust to controls for age-related and gender-related demographics and macroeconomic factors, and supports the conclusion of Mian and Sufi (2011) that a root cause of the recent financial crisis in the United States was excessive leverage among households.

7. Conclusion

This study investigates the delinquent behavior of credit card borrowers using a unique Korean account-level dataset. We define three delinquency states (i.e., no delinquency, delinquency and serious delinquency) and then test the probabilities of transitions between these states. Using panel data, observed quarterly, we find that changes in borrower characteristics during the loan period affect the transition probabilities, as do loan characteristics, the borrower's characteristics at the time of account application, and macroeconomic variables.

Several account characteristics indicate a higher delinquency risk. Seasoned accounts and accounts with larger loan amounts and shorter loan periods have a higher delinquency risk. Borrower characteristics also affect the delinquency risk. Women have a lower risk, whereas borrowers under the age of 30 have a higher delinquency risk. Borrowers in non-Seoul areas are at higher risk than

[7] While Sullivan et al. (2001) identify demographic changes as the cause of the increase in personal bankruptcy in the 1990s, Livshits et al. (2010) find that such changes contributed very little to the increase. Our results show that changes in the age composition of the borrower pool have led to an increase in the serious delinquency rate.

those in the Seoul area. A borrower's delinquency risk is also affected by his/her job status, credit risk, macroeconomic status, behavior after applying for an account, change in credit rating and DTI ratio after applying for an account, and change in job status. A higher DTI ratio and a larger remaining loan balance increase the risk of delinquency. Some accounts that are in the delinquency state recover, while others become seriously delinquent. These transition probabilities are also affected by account characteristics. Accounts with shorter loan periods are more likely to become seriously delinquent, whereas those with longer loan periods are more likely to recover from delinquency. Credit ratings and DTI ratios affect the transition probabilities. A larger current loan balance leads to a higher delinquency risk, whereas a smaller balance increases the likelihood of recovery. Extremely delinquent accounts (i.e., the default state) can recover after loan repayment. Accounts that are more likely to recover have similar characteristics: larger loan amounts, longer loan periods and smaller remaining loan balances. Borrowers between 30 and 45 years of age who live in the Seoul area are more likely to recover from delinquency, as are those with a higher credit rating and a lower DTI ratio. Favorable macroeconomic conditions increase this probability as well.

Our study contributes to the literature in several ways. Using account-level credit card loan data, we establish and investigate a model to measure delinquency risk, and then estimate the transition probabilities between delinquency states. The empirical results show how behaviors differ for borrowers in each delinquency state. Our study also has significant economic implications. First, the factors that cause delinquency differ from those that prolong delinquency and cause serious delinquency. Women and salaried employees are generally considered low-risk borrowers. However, our empirical results show that if such borrowers are in the delinquency state, they are just as likely to transit to the serious delinquency state. Second, the recent increase in the number of seriously delinquent accounts can be attributed to an increase in the number of borrowers who are older than 45. These borrowers are more likely to become seriously delinquent if they are in the delinquency state, but have a lower delinquency risk otherwise. Third, the recent increase in the number of accounts in the delinquency state can be attributed to macroeconomic variables. The difference between GDP growth and consumption growth has narrowed in recent years, which may be causing an increase in delinquency. Fourth, the DTI ratio has a profound effect on the likelihood of a transition between delinquency states. The DTI ratio is a major factor affecting recovery and delinquency, and this result is robust to controls for demographic and macroeconomic factors. Mian and Sufi (2011) also point out that household leverage played a significant role in the recent financial crisis.

There are a number of limitations to our study. For example, by including account-level data, we cannot identify holders of multiple debts. Other issues, such as debt restructuring, also need to be addressed, as in Athreya et al. (2018) and Kovrijnykh and Livshits (2017). However, in a broader sense, this study contributes to the literature by improving the understanding of unsecured debts, especially in terms of the behavior of delinquent borrowers. We will examine these issues in future work.

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